

General Introduction and Conceptual Framework of Regenerative Hydrologic Landscape Engineering

LDI Methodology – Computational Design of Infiltration Lines

1 Introduction

Water is the main structuring factor of the landscape. The way it moves across the Earth's surface largely determines the evolution of soils, the distribution of vegetation, water availability, agricultural productivity, and the stability of ecosystems.

For centuries, much of agricultural and hydraulic intervention was aimed at evacuating water as quickly as possible through channels, drainage systems, and conveyance works. This approach was suitable for certain contexts, but in many territories it also accelerated surface runoff, increased erosion, reduced infiltration, decreased groundwater recharge, and promoted the progressive degradation of the soil.

Growing climatic variability, the increasing frequency of extreme events, and the need to restore the hydrological functionality of landscapes make it necessary to evolve toward strategies that not only control water, but also allow managing its persistence, redistribution, and infiltration in a way compatible with natural processes.

This methodology arises in response to that need.

2 From hydraulic engineering to regenerative hydrologic engineering

Traditionally, hydraulic engineering has aimed to safely convey water through structures designed to transport flow rates.

Regenerative Hydrological Landscape Engineering proposes a complementary approach.

Instead of asking:

How to evacuate the water?

raises a different question:

How can we make the water stay within the landscape longer without causing damage?

This change completely modifies the way on-site interventions are designed.

The objective is no longer only to control the flow and becomes improving the landscape's overall hydrological performance.

3 The Landscape as a Living Hydraulic System

The methodology considers that a landscape is not a simple topographic surface.

It is a dynamic system composed of interconnected elements:

- relief;
- soil;
- vegetation;
- water;
- climate;
- biological activity;
- human activity.

Each of these components modifies the behavior of the others.

When any of them changes, the system's hydrological dynamics also change.

For this reason, the Infiltration Lines (LDI) should not be understood as isolated works, but as elements of a living hydraulic infrastructure integrated into the natural functioning of the territory.

4 The Movement of Water as a Design Variable

The methodology proposes an important conceptual change.

No trenches are designed.

No curves are designed.

No terraces are designed.

What is truly designed is the future behavior of the water.

Every design decision seeks to modify one or more of the following variables:

- flow velocity;
- direction of movement;
- residence time;
- Depth of infiltration;
- Spatial distribution of moisture;
- Hydrological connectivity;
- Soil stability.

The lines constitute only the mechanism by which that objective is achieved.

5 The Concept of Residence Time

One of the key concepts of this methodology is the **Water Residence Time**.

It is defined as:

The interval during which the water remains within the system before leaving the study area through runoff, deep drainage, or evaporation.

In general terms:

Longer residence time implies:

- greater infiltration;
- increased soil profile recharge;
- lower runoff velocity;

- less erosion;
- greater water availability for vegetation.

The LDI methodology considers this concept one of the main design objectives.

6 Hydrological Connectivity

Another fundamental concept is hydrological connectivity.

It is understood as the degree of continuity that allows water to move from higher areas to the system's outlet points.

High connectivity promotes:

- rapid runoff peaks;
- erosion;
- sediment transport;
- nutrient loss.

Excessively low connectivity can also cause drainage problems.

The methodology seeks a balance between both extremes.

7 Living Hydrological Infrastructure

The methodology introduces the concept of **Living Hydrological Infrastructure (LHI)**.

It is defined as:

The set of natural and designed elements that regulate the movement, storage, infiltration, and redistribution of water within the landscape.

LHI includes:

- soil;
- roots;
- Organic matter;
- Vegetation;
- Natural drains;

- infiltration lines;
- terraces;
- wetlands;
- biological corridors.

From this perspective, an Infiltration Line is one more component within a functional hydrological network.

8 Hydrological Resilience

Hydrological resilience can be defined as a landscape's ability to maintain its functioning in the face of disturbances such as:

- extreme rainfall;
- prolonged droughts;
- wildfires;
- land use changes;
- compaction;
- exceptional weather events.

Greater resilience implies:

- less erosion;
- faster recovery;
- greater soil stability;
- better use of water.

The LDI design aims to increase this resilience.

9 Data-Driven Engineering

The methodology is based on quantifiable evidence.

Every decision must be supported by objective information.

Among the main sources are:

- Digital Terrain Models;
- soil mapping;
- satellite images;

- climatic data;
- humidity sensors;
- GNSS surveys;
- field observations.

The design is no longer based exclusively on the operator's visual experience; instead, it becomes a verifiable and reproducible process.

10 From Empirical Design to Computational Design

For decades, the mapping of infiltration lines depended mainly on:

- observation of the terrain;
- visual interpretation;
- the designer's experience.

The LDI methodology proposes a different approach.

The design is developed through:

- GIS analysis;
- mathematical models;
- multi-criteria optimization;
- hydrological simulation;
- artificial intelligence;
- iterative evaluation.

The designer's experience continues to be essential, but it is complemented by quantitative tools.

11 The Infiltration Line as an Engineering Object

In this methodology, an Infiltration Line is no longer just a simple map polyline.

Each line has its own attributes:

- geometry;
- slope;
- length;
- radius;
- taxable area;
- hydraulic function;
- performance indicators;
- monitoring history.

Each one can be evaluated, modified, and validated independently.

12 The Methodology as an Adaptive System

The methodology does not assume that there is a perfect design.

Every project is an approximation that continuously improves through:

- field observations;
- monitoring;
- new images;
- new models;
- feedback.

The knowledge generated by a project is used to improve the following.

13 Scope of the Methodology

The methodology can be applied in:

- regenerative agriculture;
- sustainable livestock farming;
- silvopastoral systems;
- ecological restoration;
- watershed management;
- reforestation;
- soil conservation;
- recovery of degraded lands;
- green infrastructure.

Its modular structure allows new components to be incorporated as Earth observation technologies and hydrological models evolve.

14 Scientific principles

The methodology integrates knowledge derived from:

- hydrology;
- hydraulics;
- geomorphology;
- edaphology;
- climatology;
- landscape ecology;
- conservation agriculture;
- geographic information systems;
- mathematical optimization;
- artificial intelligence.

The combination of these disciplines makes it possible to approach the design of the LDI from a systems perspective.

15 General objective

To develop a scientific, reproducible, and automatable methodology to design Infiltration Line networks capable of improving the hydrological behavior of the landscape by increasing infiltration, redistributing water, and reducing erosion.

16 Specific Objectives

The methodology seeks to:

- increase water infiltration;
- reduce surface runoff;
- increase the water residence time;
- improve the spatial distribution of humidity;
- reduce erosion;

- Optimize agricultural mechanization;
 - Facilitate automation of the design;
 - Generate objective performance indicators;
 - Promote continuous improvement through monitoring.
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17 Vision of the methodology

The vision of the LDI methodology is to evolve from a design process into an integral platform supporting landscape hydrological management.

Under this vision, each project will have an up-to-date digital model that will integrate topographic, climatic, edaphic, and monitoring data to support decision-making before, during, and after construction.

18 Document structure

This document is organized as follows:

- **Introduction:** Introduction and conceptual framework.
 - **Chapter 1:** Scientific foundations.
 - **Chapter 2:** Design principles and basic concepts.
 - **Chapter 3:** Preparing the information and the Digital Terrain Model.
 - **Chapter 4:** Integrated methodology for the design of Infiltration Lines.
 - **Chapter 5:** Mathematical, hydrological, and hydraulic model.
 - **Chapter 6:** Computational automation and system architecture.
 - **Chapter 7:** Field implementation, monitoring, and continuous improvement.
 - **Chapter 8:** Methodological synthesis and strategic vision.
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19 Methodological Statement

The LDI methodology proposes that the design of infrastructure for water infiltration should be based on the hydrological behavior of the landscape, not solely on geometric criteria. Each design decision must be measurable, verifiable, and susceptible to improvement through field observations and computational analyses.

This approach integrates traditional technical knowledge with modern tools for spatial analysis, modeling, and optimization, aiming to develop solutions tailored to the particular conditions of each territory.

20 Final Reflection

The history of engineering has been shaped by the development of infrastructures to transport water, control flooding, and supply communities. In the current context, characterized by increased pressure on water resources and growing climate variability, there is an opportunity to complement that legacy with approaches aimed at improving the landscape's capacity to store, infiltrate, and redistribute water.

Infiltration Lines represent one of the tools available for that purpose. However, their effectiveness depends on being part of an integrated design supported by reliable information, clear technical criteria, and ongoing evaluation processes.

The methodology presented in this document proposes a framework for moving in that direction, integrating knowledge from various disciplines and promoting a reproducible, adaptable, evidence-based approach to landscape hydrological management.

Computational Methodology for the Design of Infiltration Lines (LDI)

Chapter 1. Introduction

1.1 Background

The increasing variability of climate, the rise in precipitation intensity, and the physical degradation of agricultural soils have increased the need to develop methodologies that improve water infiltration, reduce surface runoff, and enhance the resilience of productive systems.

In tropical regions with long dry seasons, such as the North Pacific of Costa Rica, most of the annual rainfall occurs during high-intensity events. When soils are compacted by machinery or livestock trampling, much of that rainfall turns into surface runoff, reducing moisture recharge in the soil profile and promoting erosive processes.

Traditional methodologies based only on contour lines or on visual interpretation of the terrain have proven useful in numerous projects, but they present limitations when precise, reproducible designs need to be developed and verified using digital tools.

Current availability of high-resolution digital terrain models, GNSS/RTK systems, drones, LiDAR sensors, and GIS platforms makes it possible to evolve toward more objective and quantifiable design procedures.

The methodology proposed in this document seeks to integrate these advances into a systematic procedure for the design of Infiltration Lines (LDI), aimed at maximizing infiltration, reducing the connectivity of surface flow, and facilitating mechanized execution of agricultural operations.

1.2 Evolution of the Concept

The design of infiltration lines has its background in various approaches to water management and soil conservation, including:

- contour farming;
- Keyline system of P. A. Yeomans;
- regenerative subsoiling;
- water harvesting (Rainwater Harvesting);

- regenerative agriculture;
- silvopastoral systems;
- hydrological restoration of watersheds.

Although these approaches share the goal of improving the distribution of water in the landscape, they differ in their design criteria, application scales, and available tools.

The LDI methodology is proposed as a computational evolution of these concepts, leveraging digital terrain models, GIS algorithms, and hydrological analysis tools to generate verifiable and reproducible designs.

1.3 General objectives

The LDI methodology is intended to design infiltration line networks that:

- increase the effective infiltration of rainfall;
- reduce the speed of surface runoff ;
- reduce the risk of water erosion;
- promote the redistribution of moisture within the landscape;
- improve conditions for establishing agricultural, forestry, and silvopastoral systems;
- are compatible with agricultural machinery equipped with GNSS systems.

1.4 Specific objectives

The specific objectives are:

1. Establish a reproducible procedure for the design of infiltration lines.
2. Integrate topographic, hydrological, and geomorphological information into a single workflow.
3. Define quantitative criteria for selecting main lines and generating infiltration patterns.
4. Incorporate hydraulic verification procedures before execution in the field.
5. Facilitate automation through GIS software and programming tools.
6. Enable continuous validation through field observations and subsequent monitoring.

1.5 Scope

The methodology is mainly oriented toward:

- degraded pastures;
- livestock farming systems;
- rainfed agriculture;
- agroforestry;
- silvopastoral systems;
- hydrological restoration;
- management of agricultural watersheds.

It can be adapted to different scales, from experimental plots to entire farms, as long as suitable topographic information is available.

1.6 Fundamental Principles

The LDI design is based on eight principles.

Principle 1. Relief Controls the Movement of Water

Every design decision must begin with a geomorphological analysis of the terrain.

Principle 2. Water must be slowed down before being conveyed

The priority is to increase the water's residence time on the ground to promote its infiltration.

Principle 3. Each Line Must Fulfill a Defined Function

The lines can be classified according to their main function:

- infiltration;
- redistribution;
- controlled conveyance;
- protection;
- support for agroforestry systems.

Principle 4. There Is No Universal Spacing

The distance between lines depends on multiple factors:

- slope;
- infiltration;
- texture;

- vegetation cover;
- rainfall intensity;
- soil storage capacity;
- type of machinery.

Principle 5. Geometry Must Be Validated Hydraulically

A visually well-ordered design does not guarantee good hydrological performance. All lines must be checked using quantitative indicators.

Principle 6. The digital terrain model is the basis of the design

The quality of the DTM directly determines the quality of the final result.

Principle 7. Automation does not replace technical judgment

Algorithms generate design proposals, but the final decision lies with the designer.

Principle 8. The Methodology Is Iterative

The design must be updated with the information obtained during the execution and monitoring of rainfall events.

1.7 Definitions

Infiltration Line (LDI): A line traced on the ground with the primary objective of promoting infiltration of rainwater, reducing surface runoff, and improving the spatial distribution of moisture.

Main Line: A reference line used to generate a family of infiltration lines through controlled shifts.

Geomorphological Unit: A sector of land with homogeneous slope, curvature, drainage, and hydrological behavior characteristics.

Longitudinal Slope: Elevation variation along an infiltration line.

Cross Slope: Slope of the terrain perpendicular to the direction of the line.

Contributing Drainage Area: The surface whose potential runoff contributes to a given line.

Hydrological Connectivity:The degree to which water can move continuously through the landscape.

1.8 Overall Flow of the Methodology

The LDI methodology is developed in nine stages:

1. Collection of baseline information.
2. Preparation of the digital terrain model.
3. Geomorphological analysis.
4. Delimitation of design units.
5. Definition of master lines.
6. Automatic generation of lines.
7. Hydraulic assessment.
8. Design settings.
9. Field validation.

Chapter 2. Scientific Foundations of the Methodology for Designing Infiltration Lines (IDL)

2.1 Introduction

Any methodology for designing Infiltration Lines (IDL) must be based on verifiable scientific principles and not solely on empirical or geometrical criteria. The spatial arrangement of the lines modifies the interaction among precipitation, infiltration, surface runoff, temporary storage of water, and the redistribution of moisture within the landscape.

The main objective is not to convey water from one point to another, but to favorably modify its hydrological behavior to increase its residence time on the ground, promote its infiltration, and reduce the energy available to generate erosion.

From this perspective, IDL constitute a distributed hydrological infrastructure that acts on the physical processes that regulate the water cycle at the surface of the earth.

2.2 Hydrological Cycle at the Farm Scale

Although the hydrological cycle is a global process, the design of the IDL focuses on the paddock scale, where management decisions directly modify the destination of rainfall water.

During a precipitation event, the total volume of water can be represented by the balance:

$$P = I + R + ET + \Delta S$$

where:

- **P**= precipitation.
- **I**= infiltration.
- **R**= surface runoff.

- ET = evapotranspiration.
- ΔS = change in water storage in the soil.

From a regenerative design perspective, the objective is to simultaneously increase I and ΔS , reducing the fraction corresponding to surface runoff without creating problems of waterlogging or ground instability.

2.3 Infiltration as a Fundamental Process

Infiltration is the process by which water passes through the soil surface and enters the profile.

Its rate depends on the interaction between physical properties of the soil, vegetative cover, initial moisture content, profile structure, and characteristics of the rainfall event.

Under ideal conditions:

- the rainfall intensity is less than the infiltration capacity;
- practically all the water enters the soil;
- surface runoff is minimal.

When the opposite situation occurs:

- the intensity of rainfall exceeds the infiltration capacity;
- the excess begins to move over the surface;
- the risk of erosion increases rapidly.

LDIs seek to delay this moment by increasing the time available for the water to infiltrate.

2.4 Soil Compaction

One of the main problems in tropical livestock and agricultural systems is compaction.

The most frequent causes are:

- repetitive machinery traffic;
- animal trampling;
- conventional tillage;
- loss of organic matter;
- reduced biological activity.

Compaction produces:

- lower macroporosity;
- lower infiltration;
- lower aeration;
- less root growth.
- greater runoff.

The implementation of LDI through subsoiling or deep chisel plowing seeks to partially break up these compacted layers, increasing the hydraulic conductivity of the profile.

2.5 Surface Water Movement

Once runoff begins, the water responds mainly to topography.

The direction of flow depends on:

- slope;
- curvature;
- roughness;
- microrelief;
- natural obstacles.

Initially, the flow is diffuse.

Subsequently, it begins to concentrate.

Finally, the following appear:

- rills;
- gullies;
- temporary drainage channels;
- permanent channels.

LDIs seek to intervene before the flow reaches enough energy to begin these processes.

2.6 Potential energy of water

Water has potential energy due to the difference in elevation.

During its descent, this energy is transformed into speed.

The slope controls much of this process.

The steeper the slope:

- greater speed;
- greater erosive capacity;
- less time available for infiltration.

Therefore, one of the design objectives is to increase the effective path of the water over the hillside, while simultaneously reducing the effective hydraulic gradient.

2.7 Infiltration versus runoff

Historically, many designs have confused both concepts.

An infiltration line **should not behave like a drainage channel**.

Its main function is:

- to slow down;
- to infiltrate;
- to redistribute.

Not to evacuate the water quickly.

When a line begins to convey increasing flows over long distances, problems such as arise:

- longitudinal erosion;
 - cutting (incision);
 - formation of gullies;
 - loss of the regenerative effect.
-

2.8 Distributed Hydrology

The LDI methodology adopts an approach to distributed hydrology.

Instead of analyzing only a single point on the terrain, the spatial behavior of the entire surface is considered.

Each cell of the Digital Terrain Model represents a hydrological unit with its own attributes:

- elevation;
- slope;
- aspect;
- curvature;
- contributing area;
- moisture index;
- connectivity.

This allows designing different lines according to the specific conditions of each sector of the land.

2.9 Geomorphology of the Landscape

Geomorphology studies the shapes of the relief and the processes that create them.

On a farm, the following can normally be distinguished:

- watersheds;
- convex slopes;
- straight slopes;
- concave slopes;
- ravines;
- channels.

Each one has a different hydrological behavior.

Therefore, it is not recommended to apply a uniform pattern across the entire property.

2.10 Terrain Curvature

The slope describes only the inclination.

Curvature describes how that slope changes.

There are two main components.

Profile Curvature

Controls:

- flow acceleration;
- deceleration;
- potential erosion.

Plan-View Curvature

Controls:

- flow convergence;
- flow divergence;
- lateral concentration.

These variables are fundamental for deciding where it is advisable to increase or decrease the line density.

2.11 Contributing Area

Each point on the ground receives water from a specific surface.

This surface is called the contributing area.

The greater it is:

- greater potential volume;
- greater hydraulic risk;
- utmost need for verification.

The automatic calculation of the contributing area is one of the pillars of the LDI computational methodology.

2.12 Hydrological Indices

Various indices derived from the DTM make it possible to characterize the potential behavior of water.

Among the most important are:

Topographic Wetness Index (TWI)

Makes it possible to identify areas with a higher probability of moisture accumulation.

Stream Power Index (SPI)

Relates slope and accumulation.

Identifies potential erosion-prone areas.

LS Factor

Represents the combined effect of length and slope used in RUSLE.

HAND

Relative elevation with respect to the nearest drainage.

Very useful for differentiating slopes and flood-prone areas.

2.13 Hydrological Connectivity

Connectivity represents how easily water can move across the landscape.

A highly connected terrain supports:

- rapid runoff peak flows;
- erosion;
- sediment transport.

The LDI aim to reduce this connectivity by creating multiple distributed infiltration points.

2.14 Surface Runoff Hydraulics

Once runoff appears, the movement can be approximated using Manning's equation:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

where:

- **V**= flow velocity;
- **n**= roughness coefficient;
- **R**= hydraulic radius;
- **S**= hydraulic slope.

Although LDIs normally do not operate as permanent channels, this equation makes it possible to estimate the potential flow behavior in extreme scenarios.

2.15 Shear Stress

Erosive capacity can be assessed by:

$$\tau = \rho g R S$$

where:

- **τ** = shear stress;
- **ρ** = density of water;
- **g**= gravity;
- **R**= hydraulic radius;
- **S**= slope.

When the stress exceeds the soil resistance, the detachment of particles begins.

One of the design objectives is to keep this value below the critical threshold.

2.16 Design Spatial Scale

Not all decisions must be made at the same scale. The methodology distinguishes four levels:

Resolution Scale	Goal
Land Plot 10–30 m	Hydrological Context
Scale 1–5 m	Overall Design
Operational 0.10–0.50 m	Line tracing
RTK execution (2–5 cm)	Field construction

This hierarchy prevents using information that is too coarse for tasks that require high precision.

2.17 Influence of the weather

The behavior of the LDI depends on the rainfall regime.

In seasonal tropical regions, the following predominate:

- intense convective storms;
- long dry periods;
- high interannual variability.

Therefore, the design should not be based solely on average precipitation, but should consider extreme events that represent the most demanding conditions for the system.

2.18 The Soil as Hydraulic Infrastructure

Traditionally, hydraulic infrastructure has been associated with civil engineering works such as canals, dams, or sewers.

The LDI methodology proposes a different approach: considering the soil as a living infrastructure capable of storing, infiltrating, and redistributing water.

The efficiency of the system depends on the interaction between:

- soil physical structure;
- roots;
- organic matter;
- biological activity;
- vegetative cover;
- grazing or crop management.

Infiltration lines are the organizing element of that living infrastructure, but their performance is conditioned by the quality of the soil that supports them.

2.19 Conceptual Synthesis

The scientific foundations of the LDI methodology can be summarized in five principles:

1. **Relief controls the movement of water.**
2. **The soil regulates infiltration and storage.**
3. **The infiltration lines modify the landscape's hydrological connectivity.**
4. **The design must be based on quantifiable variables derived from the DTM and verified through hydrological and hydraulic analyses.**
5. **The system's effectiveness depends both on the geometry of the lines and on subsequent soil and vegetation management.**

Chapter 3. Base Information and Preparation of the Digital Terrain Model (DTM)

3.1 Introduction

The quality of the design of a network of Infiltration Lines (LDI) depends directly on the quality of the topographic information used. An advanced design algorithm cannot compensate for a Digital Terrain Model (DTM) with systematic errors, insufficient resolution, or processing artifacts.

Consequently, the first step of the methodology consists of building a digital model that faithfully represents the surface of the bare ground, free of vegetation, buildings, or other elements that alter hydrological analysis.

The accuracy of the DTM determines all subsequent stages: slope calculation, delimitation of drainage areas, identification of geomorphological units, generation of lines, and hydraulic verification.

3.2 Working Scales

The design of LDI can be developed at different scales. The DTM resolution must be selected according to the project objective.

Project scale Project size Recommended resolution		
Regional	>10 000 ha	10–30 m
Basin	500–10 000 ha	2–10 m
Farm	10–500 ha	0.5–2 m
Operational design	<100 ha	0.10–0.50 m
Reframing RTK Field		2–5 cm

For projects where the lines will be separated by 3 to 5 m, a 10 m DTM is insufficient. The methodology recommends working, whenever possible, with resolutions equal to or less than 1 m.

3.3 Sources of Information

The methodology supports multiple data sources, provided they meet the minimum accuracy requirements.

3.3.1 LiDAR

It is the preferred source.

Advantages:

- high vertical accuracy;
- high point density;
- the ability to obtain bare ground;
- excellent representation of microrelief.

Limitations:

- High cost;
 - Limited availability in some countries;
 - Requires specialized processing.
-

3.3.2 Drone Photogrammetry

It currently constitutes one of the best alternatives for property development projects.

Advantages:

- High spatial resolution;
- Relatively low cost;
- Fast updates;
- Integration with RTK/PPK.

Limitations:

- tall vegetation makes it difficult to obtain the actual terrain;
 - requires photogrammetric processing.
-

3.3.3 RTK Surveys

Very useful for:

- critical areas;
- validation;
- re-planning;
- DTM calibration.

They do not replace a complete terrain survey when total coverage is required.

3.3.4 Global Models

Among them:

- Copernicus DEM
- ALOS AW3D
- FABDEM
- SRTM
- NASADEM

They are suitable for regional studies, but generally do not have the resolution required for detailed LDI design.

3.4 Reference System

All information must use the same coordinate reference system.

It is recommended:

- projected system;
- units in meters;
- official datum of the country;
- a consistent geoidal model for heights.

Geographic coordinates and measurements must not be mixed during design.

3.5 Required Accuracy

Vertical accuracy is more important than horizontal. Recommended values:

Parameter	Recommendation
Vertical RMSE Error	<0.10 m
Horizontal Error	<0.20 m
RTK Error	2–5 cm
Separation between RTK points	5–20 m

On gentle slopes, small vertical errors can completely change the meaning of surface flow.

3.6 Digital Surface Model and Digital Terrain Model

It is important to distinguish between these two concepts.

Digital Surface Model (DSM)

It represents:

- trees;
- buildings;
- fences;
- infrastructure;
- vegetation.

It is suitable for landscape studies.

It must not be used directly for hydraulic design.

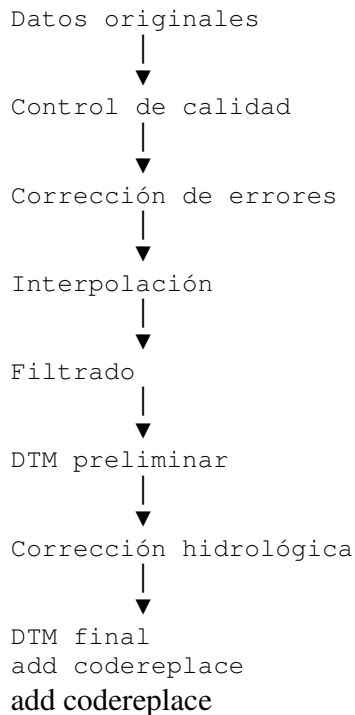
Digital Terrain Model (DTM)

It represents only the bare ground.

It is the model used by the LDI methodology.

3.7 DTM Preparation Workflow

The recommended processing comprises the following stages:



Each stage must be documented to ensure process traceability.

3.8 Initial Quality Control

Before any analysis, the following are checked:

- spatial coverage;
- resolution;
- null values;
- coordinates;
- units;
- continuity;
- obvious errors.

The design should not be started while inconsistencies exist.

3.9 Artifact Removal

DTMs usually contain:

- isolated spikes;
- artificial depressions;
- stepped surfaces;
- photogrammetric noise;
- interpolation errors.

These defects create nonexistent drainage.

Therefore, they must be removed before hydrological analysis.

3.10 Hydrological Correction

Once the DTM has been refined, the hydrologic correction is carried out.

Its objective is to ensure that surface flow can move correctly.

The usual operations include:

- removal of spurious depressions;
 - breaching of artificial barriers;
 - correction of bridges and roads;
 - drainage continuity.
-

3.11 Optimal Resolution

There is a trade-off between resolution and processing time.

Resolution	Recommended use
10 m	Regional studies
5 m	Basins
2 m	Preliminary design
1 m	Recommended design
0.5 m	Detailed design
0.10–0.20 m	High-precision layout setting out

Higher resolution does not always mean better quality if it increases the model's noise.

3.12 Topographic derivatives

Once the final DTM is obtained, the derived products are generated.

Pending

Allows you to identify:

- areas that can be mechanically processed;
 - erosion risk;
 - operational limitations.
-

Orientation (Aspect)

Determines:

- insolation;
 - evaporation;
 - differential moisture behavior.
-

Curvature

It is calculated:

- profile curvature;
 - plan curvature;
 - general curvature.
-

Contour Lines

The methodology recommends generating two

families. Design curves:

- equidistance of 0.25–0.50 m.

Presentation curves:

- equidistance of 1 m.
-

3.13 Hydrological variables

Subsequently, the following are calculated:

- flow direction;
- flow accumulation;
- contributing area;
- TWI;
- SPI;
- HAND;
- LS factor;
- drainage network.

These maps form the basis for defining design units.

3.14 Automatic delineation of drainage

Not all channels are visible in aerial images.

Hydrological analysis makes it possible to identify:

- permanent drainage;
- temporary drainage;
- swales;
- potential concentrations.

The LDI lines must follow these natural structures.

3.15 Delimitation of Exclusion Zones

Before the design, areas are identified where infiltration lines must not be built. These include:

- causes;
- wetlands;
- springs;
- unstable slopes;
- gullies;
- infrastructure;
- main roads;
- buildings;
- protected areas.

These areas are incorporated as mandatory design algorithm constraints.

3.16 Geomorphological Classification

The terrain is divided into relatively homogeneous units.

Examples:

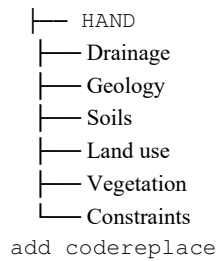
- convex slopes;
- straight slopes;
- concave slopes;
- valley bottoms;
- watersheds;
- terraces.

Each unit will receive a different treatment during the design.

3.17 Multi-Layer Integration

The LDI methodology works through the superposition of thematic layers.

DTM
 |
 Pending
 Curvature
 Aspect
 |— Flow Direction |—
 Flow Accumulation
 TWI
 SPI



The combination of these layers makes it possible to make informed decisions for each section of the land.

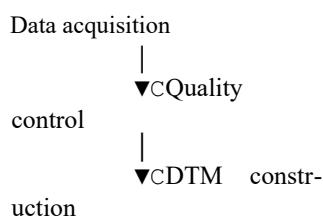
3.18 Minimum Acceptance Criteria for the DTM

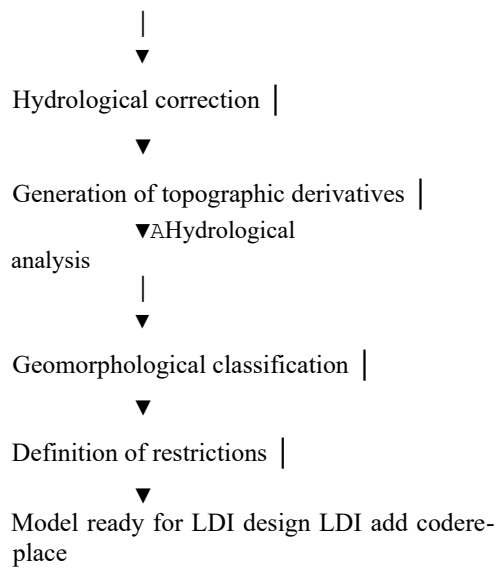
Before starting the design of the IDIs, the model must comply at minimum with:

Variable	Criterion
Coverage	100 % of the work area
Coordinate system	Verified
Resolution	Compatible with the spacing of the LDIs
Vertical errors	Within tolerance
Hydrological continuity	Verified
Artifacts	Eliminated
Drainage outlets	Reviewed
Contour lines	Generated
Hydrological variables	Calculated

If any of these criteria is not met, the process must stop until the problem is corrected.

3.19 General Preparation Flow





3.20 Chapter Conclusions

The Digital Terrain Model constitutes the foundation on which the entire LDI methodology is developed. The accuracy of the DTM determines the reliability of subsequent analyses and, therefore, the quality of the final design.

This chapter establishes a standardized preparation flow that ensures the digital terrain faithfully represents the actual hydrological conditions before starting the layout of the lines.

Chapter 4. Comprehensive Methodology for Designing Infiltration Lines (LDI)

4.1 Introduction

The previous chapters established the scientific foundations and the preparation of geospatial information. This chapter is the core of the methodology, describing the complete procedure to design an Infiltration Lines (LDI) network using high-resolution topographic data, hydrological analysis, and engineering criteria.

Unlike approaches based solely on contour lines or on the automatic generation of parallel lines, the LDI methodology proposes an iterative process in which each line is evaluated as an independent hydraulic entity. The goal is to ensure that the network operates as an integrated system capable of increasing infiltration, reducing hydrological connectivity, and maintaining operational viability for agricultural machinery.

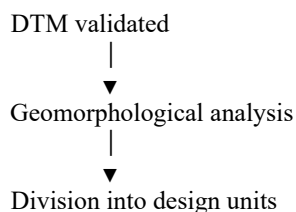
4.2 Design Philosophy

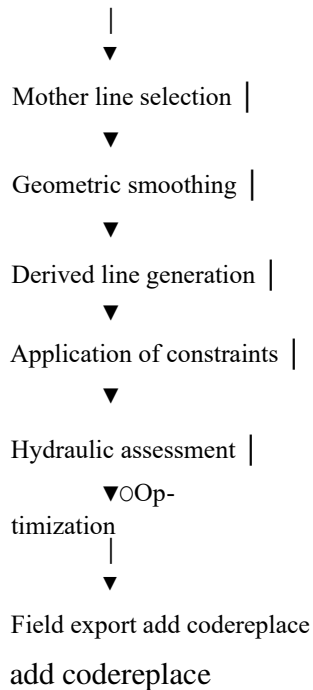
The methodology is based on four operating principles:

1. **The terrain defines the design; the designer interprets the terrain.**
 2. **Each line must fulfill a specific hydrological function.**
 3. **Geometry must be verified using hydraulic and operational criteria.**
 4. **The design is refined through successive iterations until achieving acceptable hydrological behavior.**
-

4.3 General Process Flow

The complete procedure can be represented by the following flow:





4.4 Delimitation of Design Units (DUs)

Before drawing any line, the property must be divided into **Design Units (DUs)**.

A Design Unit is a portion of land that shows a relatively homogeneous hydrological behavior and that can be treated using the same set of design criteria.

The DUs constitute the basic unit of work of the methodology.

4.4.1 Objectives

- avoid applying a uniform design to heterogeneous terrains;
- facilitate local optimization;
- adapt the spacing and orientation of the lines;
- improve mechanization.

4.4.2 Variables Used

The UD are defined by integrating:

- pending;
 - curvature;
 - orientation (aspect);
 - flow accumulation;
 - soil type;
 - effective depth;
 - land use;
 - drainage;
 - obstacles.
-

4.5 Functional classification of lines

One of the original contributions of the LDI methodology is to classify each line according to its hydraulic function.

Type A – Infiltration lines

Goal:

Increase infiltration.

Characteristics:

- very low slope;
 - maximum continuity;
 - high density.
-

Type B – Redistribution lines

Goal:

Slowly transfer moisture from damp areas to drier areas.

Type C – Dissipation Lines

Objective:

Reduce hydraulic energy before critical zones.

Type D – Agroforestry lines

Objective:

To serve as an axis for:

- trees;
- living fences;
- biological corridors;
- silvopastoral systems.

Type E – Exclusion lines

They correspond to:

- drainage systems;
- broken
- springs;
- wetlands;
- slopes.

They cannot be modified.

4.6 Selection of the Main Line

Each Design Unit must have at least one Main Line.

The Main Line represents the geometric pattern that will control the rest of the design.

Selection Criteria

It must meet simultaneously:

✓ topographical continuity

✓ stable slope

- ✓ Suitable radio
 - ✓ Low accumulation
 - ✓ Ease of mechanization
 - ✓ Absence of drainage ditches
-

4.7 Obtaining the Main Line

It can be obtained by:

Method 1

Manual digitization.

Method 2

From contour lines.

Method 3

Using an automatic algorithm.

Method 4

Combination of AI and manual review.

The methodology always recommends the final technical review.

4.8 Smoothing

The line initially obtained usually contains:

- unnecessary vertices;
- breaks;
- oscillations.

Smoothing must preserve:

- slope;
- continuity;
- overall geometry.

It must not modify hydraulic behavior.

4.9 Minimum geometric parameters

Parameter	Recommended value
Minimum radius	10–15 m
Minimum length	20–30 m
Maximum continuous length	150–250 m*
Initial spacing	3–5 m
Longitudinal slope	0–0.5 %

(*) Dependent on hydraulic input.

4.10 Generation of derived lines

Once the Mother Line has been validated, parallel lines are generated using controlled offsets.

It is important to note that these lines **are not simple geometric offsets**.

Each line must then be adjusted to the terrain.

4.11 Variable Spacing

One of the proposed changes with respect to the original document is to abandon the concept of fixed spacing.

Instead of always using 3.5 meters, the methodology proposes:

$$S = f(P, T, I, U, A)$$

where

S = spacing

P = slope

T = texture

I = infiltration

U = land use

A = contributing area

This makes it possible to automatically increase or decrease the line density according to local conditions.

4.12 Adjustment for Slope

Slopes with a higher incline require:

- shorter length;
- greater hydraulic control;
- wider radii.

Flat sections allow:

- longer lines;
- greater spacing.

4.13 Adjustment for Accumulation

Lines must never pass through areas with high flow accumulation without a specific assessment.

The algorithm can:

- cut the line;
 - modify its trajectory;
 - turn it into a dissipation line;
 - exclude the sector.
-

4.14 Adjustment for curvature

Convex and concave slopes require different behaviors.

In convexities:

- less separation;
- greater continuity.

In concavities:

- more caution;
 - input control.
-

4.15 Densification

Each line must be converted into a regularly spaced sequence of points.

Recommendation:

5 m

In closed curves:

2 m

4.16 DTM Sampling

For each point, the following is extracted:

- elevation;
- slope;
- curvature;
- accumulation;
- humidity;
- soil.

This information will be used in the hydraulic assessment.

4.17 Individual Assessment

Each line stops being a simple polyline and becomes an object with attributes.

Example:

Variable
Length
Average slope
Maximum slope
Minimum radius
Catchment area
Maximum accumulation
Erosive risk
Line type
Status

This constitutes one of the main methodological contributions of the LDI.

4.18 Scoring System

It is proposed to assign a Line Quality Index (LCI).

Example:

Parameter	Weight
Slope	25
Accumulation	25

Weight Parameter

Curvature	15
Radius	15
Length	10
Ground	10

Max ICL = 100

Classification:

90–100 Excellent

80–89 Good

70–79 Acceptable

60–69 Review

<60 Redesign

4.19 Iterative Optimization

The algorithm runs several iterations.

In each cycle:

1.

Evaluates.

↓

2.

Detects problems.

↓

3.

Corrects.

↓

4.

Reevaluate.

↓

5.

Accept or reject.

The methodology stops being linear to become an iterative optimization process.

4.20 Mandatory Restrictions

The lines must comply with:

- drainage;
- wetlands;
- infrastructure;
- main fences;
- roads;
- springs;
- protected areas.

These restrictions have absolute priority.

4.21 Multi-Criteria Optimization

The design seeks to simultaneously maximize:

- infiltration;
- mechanization;
- hydraulic stability;
- constructive simplicity.

And minimize:

- erosion;

- earthwork;
 - unnecessary length;
 - drainage crossings.
-

4.22 Visual Validation

Before exporting the design, it is recommended to carry out a review by means of:

- shading of the relief;
- longitudinal profiles;
- 3D visualization;
- orthophotos;
- virtual tour.

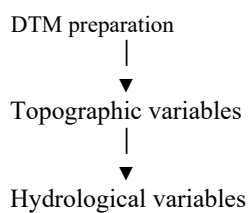
Many geometric errors are more easily detected through visual inspection than through statistics.

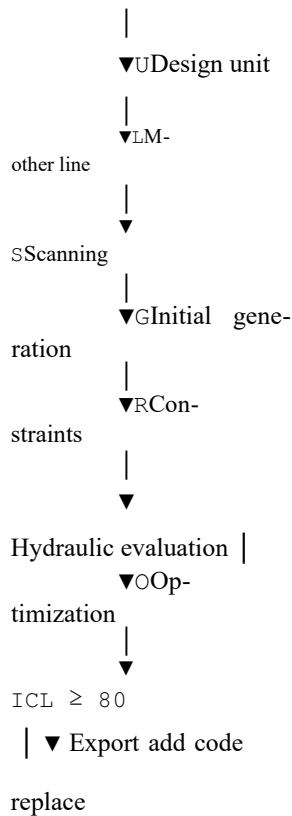
4.23 Field Preparation

Once the lines are approved:

- automatic numbering;
 - identification by Design Unit;
 - coordinate table;
 - export to GeoPackage;
 - export to GeoJSON;
 - export to SHP;
 - export for GNSS;
 - PDF plan generation.
-

4.24 Complete Design Flow





4.25 Proposed Methodological Innovations

This chapter introduces several improvements over the original material:

1. **Design Units (UD):**the land is divided into homogeneous sectors before layout, avoiding applying a single pattern to the entire property.
2. **Functional classification of the lines:**each LDI is designed with an explicit function (infiltration, redistribution, dissipation, agroforestry, or exclusion), rather than assuming that all of them serve the same purpose.
3. **Variable spacing:**the separation between lines is no longer a fixed value (e.g., 3.5 m) and instead depends on the slope, soil, infiltration, land use, and the contributing area.
4. **Line Quality Index (ICL):**each line is assessed with a quantitative score before being accepted for construction, turning the design into an objective and repeatable process.
5. **Iterative optimization:**the design is automatically corrected through successive evaluations, bringing the methodology closer to an engineering approach assisted by GIS and artificial intelligence.

Chapter 5. Mathematical, Hydrological, and Hydraulic Model of the LDI Methodology

5.1 Introduction

The previous chapters described the scientific foundations and the design procedure. This chapter develops the mathematical framework that supports the LDI methodology.

The purpose is not to build a fully distributed hydrological model like SWAT, MIKE SHE, or ParFlow, but to define a set of equations, indicators, and objective functions that allow for a quantitative assessment of the performance of each Infiltration Line (LDI) before it is constructed.

The philosophy behind the methodology is that **each infiltration line must behave like a verifiable hydraulic element**, with measurable attributes and objective acceptance criteria.

5.2 Model scales

The model operates at four hierarchical levels.

Level Analysis unit		Result
Cell	DTM pixel	Topographic variables
Line	Each LDI	Hydraulic indicators
Combined	Design Unit of line	Local optimization
perpty	Complete network	Global performance

This structure facilitates the analysis of both individual components and the integrated behavior of the entire network.

5.3 Fundamental Variables

The methodology uses the following primary variables.

Symbol	Variable	Unit
z	Elevation	m
S	Slope	m/m
A	Contributing area	m ²
L	Line length	m
R_h	Hydraulic radius	m
Q	Flow rate	m ³ /s
V	Velocity	m/s
I_f	Infiltration	mm/h
P	Precipitation	mm
ET	Evapotranspiration	mm
S	Soil moisture	%
TWI	Topographic wetness index	-
SPI	Flow power index	-

5.4 Water Balance

For each Design Unit, the balance is established:

$$P = I + R + ET + \Delta S$$

where:

- **P**= precipitation;
- **I**= infiltration;
- **R**= runoff;
- **ET**= evapotranspiration;
- **ΔS**= soil storage.

The purpose of the methodology is to maximize:

$$I + \Delta S$$

and minimize:

$$R$$

without compromising the stability of the ground.

5.5 Slope

The local slope is calculated using the gradient of the DTM:

$$S = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}$$

This variable controls:

- velocity;
 - energy;
 - infiltration;
 - erosion.
-

5.6 Longitudinal Slope of a Line

Each LDI has its own slope.

Between two consecutive points:

$$i = \frac{\Delta z}{\Delta L}$$

where:

$$\Delta z = z_2 - z_1$$

and

$$\Delta L$$

is the horizontal distance.

This gradient is one of the main acceptance indicators.

5.7 Maximum gradient

In addition to the average gradient, the following is calculated:

$$i_{max}$$

Which corresponds to the highest value found along the line.

A line may have an acceptable average gradient and still include a short segment that is highly erosive.

5.8 Curvature

Curvature is obtained from the second derivatives of the DTM.

There are:

- profile curvature;
- plan curvature.

These variables make it possible to identify:

- divergent zones;
 - convergent zones;
 - flow acceleration.
-

5.9 Contributing Area

For each point:

$$A = f(\text{Flow Direction})$$

This variable represents the surface that contributes runoff to that point. The larger the

contributing area:

- greater potential discharge;
 - greater need for control.
-

5.10 Topographic Wetness Index

The TWI is calculated by:

$$TWI = \ln\left(\frac{A}{\tan \beta}\right)$$

Where:

$$\beta$$

Is the slope.

High values indicate:

- greater probability of moisture;
 - greater water retention.
-

5.11 Flow Power Index

$$SPI = A \tan \beta$$

It represents the potential energy of surface runoff.

High values indicate:

- greater erosive capacity;
 - higher priority for assessment.
-

5.12 Hydraulic Length

Not all of the geometric length contributes to the flow.

It is defined as:

$$L_h$$

the hydraulically connected length.

A long line can behave hydraulically like several independent lines if there are natural or designed interruptions.

5.13 Residence Time

One of the main objectives is to increase the time during which the water remains on the ground.

It is approximated by:

$$T_r = \frac{L}{V}$$

The greater it is:

- greater potential infiltration;
 - lower runoff peak.
-

5.14 Velocity

Velocity can be estimated using Manning:

$$V = \frac{1}{n} R_h^{2/3} S^{1/2}$$

where:

n represents roughness.

Vegetation cover increases roughness and decreases speed.

5.15 Shear stress

$$\tau = \rho g R_h S$$

Yes

$$\tau > \tau_c$$

Erosion begins.

The design should aim to ensure that:

$$\tau < \tau_c$$

For the given soil.

5.16 Hydraulic power

It can also be estimated:

$$\Omega = \gamma Q S$$

Where:

γ represents the specific weight of water.

This parameter is useful for comparing design scenarios.

5.17 Infiltration Capacity

It can be represented by Horton:

$$f(t) = f_c + (f_0 - f_c)e^{-kt}$$

where:

- f_0 = initial infiltration;
- f_c = steady infiltration;
- k = decay constant.

The methodology makes it possible to replace this model with others (Green-Ampt, Philip, etc.) depending on data availability.

5.18 Index of Hydrological Connectivity (ICH)

It is proposed to incorporate a new indicator:

$$ICH = \frac{\sum L_c}{A_p}$$

where:

- L_c = length hydraulically connected;
- A_p = area of the property.

High values indicate:

- rapid transfer of water;
- higher erosive risk.

The purpose of the LDI is to reduce this index.

5.19 Infiltration Capacity Index (ICI)

New proposed indicator.

$$ICI = f(P, S, T, O, C, S)$$

where they come into play:

- precipitation;
- initial moisture;
- texture;
- organic matter;
- ground cover;
- slope.

This index will allow automatic adjustment of the spacing between lines.

5.20 Objective Function

The methodology is formulated as an optimization problem. It

is desired to maximize:

$$F = w_1 I + w_2 T_r + w_3 S - w_4 R - w_5 E$$

where:

- I = infiltration;
- Tr = residence time;
- SM = moisture;
- R = runoff;
- E = erosion.

The coefficients:

$$w_i$$

They allow adapting the methodology to different objectives.

5.21 Restrictions

The model is subject to restrictions.

Examples:

Pending:

$$S < S_{max}$$

Radius:

$$R > R_{min}$$

Separation:

$$D > D_{min}$$

Buffer:

$$B > B_{min}$$

These constraints may vary depending on the project.

5.22 Multi-Criteria Optimization

The algorithm solves simultaneously:

Maximize

- infiltration;
- moisture;
- mechanization;
- stability.

Minimize

- erosion;

- earthmoving;
- connectivity;
- unnecessary length.

5.23 Global LDI Index (IG-LDI)

An integral indicator is proposed to evaluate the overall design.

$$IG = 0.20(ICL) + 0.20(ICI) + 0.15(ICH) + 0.15(HM) + 0.15(ME) + 0.15(CQ)$$

where:

ICL = Line Quality Index.

ICI = Infiltration Capacity Index.

ICH = Connectivity Index.

HM = Expected moisture.

ME = Mechanization.

CQ = Geometric quality.

Proposed scale:

IG-LDI Classification

>90	Excellent
80–90	Very good
70–80	Good
60–70	Acceptable
<60	Redesign

5.24 Scenario Evaluation

The model allows comparing design alternatives.

Example:

Scenario A

- separation 3 m.

Scenario B

- separation 4 m.

Scenario C

- variable separation.

Each alternative is assessed using the same set of indicators.

5.25 Sensitivity

All methodology must analyze how results change when variables change.

Critical variables:

- rainfall intensity;
- slope;
- infiltration;
- roughness;
- initial moisture;
- DTM resolution.

This makes it possible to identify which parameters influence the system's behavior the most.

5.26 Calibration

The parameters must be adjusted using field measurements.

Among them:

- infiltration;
- humidity;
- runoff;
- effective depth;

- compaction;
- flow speed.

The methodology assumes that the model improves continuously as more information is observed.

5.27 Validation

Validation must be carried out by comparing:

Prediction

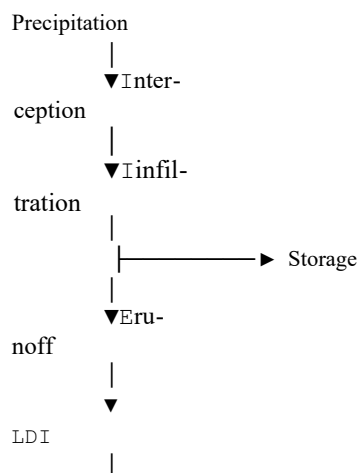
vs.

Observation.

Recommended indicators:

- measured infiltration;
- soil moisture;
- photographs after rain;
- absence of erosion;
- persistence of the lines;
- plant productivity.

5.28 Integrated Conceptual Model



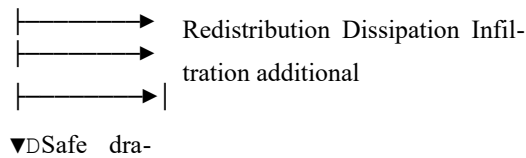


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5.29 Towards a Hydrological Digital Twin (Digital Twin)

As a future evolution of the methodology, it is proposed that each LDI project have an **Hydrological Digital Twin**, which integrates:

- the high-resolution DTM;
- the complete LDI network;
- soil moisture sensors;
- weather stations;
- satellite data (for example, surface moisture, NDVI, and precipitation);
- management records (tillage, grazing, seeding);
- periodic monitoring using drones.

This model would allow for continuously comparing the expected behavior with the observed one and adjusting the design as the system evolves.

5.30 Conclusions of the chapter

The mathematical model of the LDI methodology converts the design of infiltration lines into a quantifiable and optimizable problem. Integrating topographic, hydrological, hydraulic, and operational variables makes it possible to objectively evaluate different alternatives before implementation, reducing reliance on criteria that are exclusively empirical.

Proposed Innovations for the Final Version of the Methodology

This chapter includes several contributions that expand the focus of the original document:

1. **Mathematical formalization of the design**, expressing it as an optimization problem with objective functions and constraints, rather than a sequence of design rules.
2. **Three new integrative indicators:**
 - **ICH (Hydrological Connectivity Index)** to measure the degree of connectivity of surface flow.
 - **ICI (Infiltration Capacity Index)** to adapt the design to local soil and rainfall conditions.
 - **IG-LDI (Global LDI Index)** to assess the performance of the entire line network.
3. **Continuous calibration approach**, incorporating field data to progressively improve the model.
4. **Hydrological Digital Twin Vision**, which would make it possible to integrate GIS, sensors, drones, and field observations into a dynamic decision-support platform.

Chapter 6. Automation

Computational Aspect of the LDI Methodology

GIS Architecture, Artificial Intelligence, and Parametric Design

6.1 Introduction

The main difference between the LDI methodology and traditional design methods is that the process can be automated almost entirely through Geographic Information Systems (GIS), optimization algorithms, and artificial intelligence.

The goal of automation is not to replace the designer's judgment, but to assist them in analyzing thousands of combinations that would be impossible to evaluate manually.

Automation turns the design of Infiltration Lines into a process:

- reproducible;
 - auditable;
 - quantifiable;
 - Scalable;
 - Optimizable.
-

6.2 Automation Philosophy

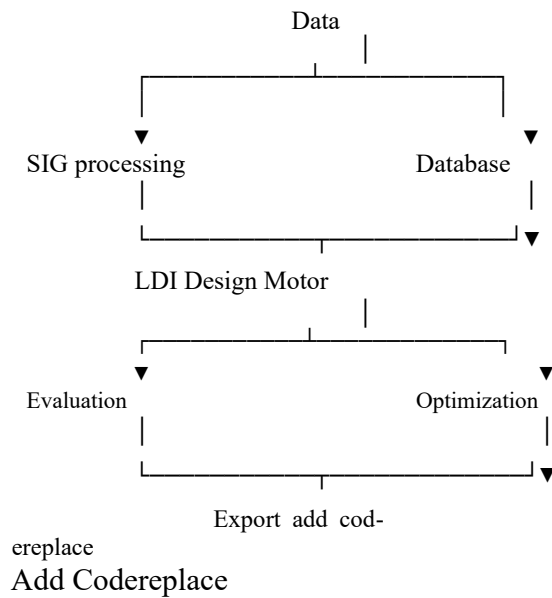
Automation follows five fundamental principles:

1. **All calculations must be reproducible.**
2. **Every parameter must be documented.**
3. **Each stage must be able to run independently.**
4. **The user can intervene at any stage of the process.**
5. **The methodology must be independent of the specific software.**

This makes it possible to implement the workflow on different platforms without modifying the design principles.

6.3 General Architecture

The Methodology Is Organized Into a Modular Architecture.



Each Module Can Evolve Without Affecting the Others.

6.4 Project Organization

It Is Recommended to Maintain a Standard Structure.

```
Proyecto_LDI/  
01_Datos_Brutos/  
02_DTM/  
03_Hidrologia/  
04_Geomorfologia/  
05_Unidades_Disenio/  
06_Linea_Madre/  
07_Lineas_LDI/  
08_Optimizacion/
```

09_Validation/

10_Outputs/

Scripts/

Documentation/

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This structure facilitates traceability and collaborative work.

6.5 Overall Processing Flow

The complete flow consists of twelve stages.

1. Data import.

↓

2. Validation.

↓

3. DTM construction.

↓

4. Hydrological correction.

↓

5. Topographic variables.

↓

6. Hydrological variables.

↓

7. Delimitation of units.

↓

8. Initial design.

↓

9. Evaluation.

↓

10. Optimization.

↓

11. Validation.

↓

12. Export.

6.6 Functional Modules

Automation is divided into independent modules.

Module 1

Import.

Functions:

- load data;
 - check formats;
 - check projection;
 - validate resolution.
-

Module 2

DTM.

Functions:

- generate DTM;
- soften;
- remove noise;

- correct depressions.
-

Module 3

Topographic analysis.

Calculate:

- slope;
 - aspect;
 - curvatures;
 - local relief.
-

Module 4

Hydrological analysis.

Generate:

- flow direction;
 - accumulation;
 - TWI;
 - SPI;
 - HAND;
 - drainage.
-

Module 5

Geomorphology.

Automatically classify:

- slopes;
 - divides;
 - gullies;
 - terraces;
 - valley bottoms.
-

Module 6

Line generation.

Responsible for:

- mother line;
 - offsets;
 - cuts;
 - connections;
 - smoothing.
-

Module 7

Evaluation.

Calculate:

- slope;
 - radii;
 - accumulation;
 - risk;
 - ICL.
-

Module 8

Optimization.

Correct automatically:

- radii;
 - length;
 - separation;
 - slope.
-

Module 9

Export.

Produce:

- GeoPackage;
- GeoJSON;
- SHP;
- CSV;
- DXF;
- PDF.

6.7 Object-Oriented Model

Each line stops being a polyline to become an object.

Conceptual example:

LDI Class

ID

Geometry

Length

Slope

Curvature

Tributary area

Type

ICL

Status

Observations

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This approach facilitates automation, quality control, and subsequent analysis.

6.8 Internal Flow of a Line

Each line goes through the following cycle:

Create

↓

Evaluate

↓

Optimize

↓

Evaluate again

↓

Accept

↓

Export

add codereplace

If the line does not meet the criteria, return to the optimization cycle.

6.9 Rule Engine

The algorithm uses explicit rules.

Examples:

IF pending > limit →

decrease length. IF

high accumulation →

split line.

IF small radius

→ smooth.

IF it crosses a drai-

nage area → remove.

IF ICL < 80

→ redesign.

These rules can be expanded depending on the type of project.

6.10 Parametric optimization

The methodology incorporates adjustable parameters.

Examples:

- initial separation;
- maximum slope;
- minimum radius;
- maximum length;
- drainage buffer;
- line density.

Each parameter can be modified without changing the main algorithm.

6.11 Automation through artificial intelligence

AI does not replace the physical model.

Its function consists of:

- detect patterns;
- propose improvements;
- classify geomorphology;
- estimate parameters;
- prioritize alternatives.

The final decision always corresponds to the designer.

6.12 Multi-Criteria Optimization

Automation evaluates simultaneously:

Maximize:

- infiltration;
- moisture;
- continuity;
- ease of construction.

Minimize:

- erosion;
- length;
- earthwork movement;
- complexity.

6.13 Evaluation Engine

Each line automatically receives a set of indicators. Example:

Indicator	Status
Pending	✓
Curvature	✓
Taxable area	✓
TWI	✓
SPI	✓
ICL	✓
Risk	✓

The designer can quickly identify the lines that require review.

6.14 Optimization Flow

Initial design

↓

Evaluation

↓

Errors

↓

Correction

↓

New evaluation

↓

Convergence

↓

Final design

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The process continues until the established criteria are met.

6.15 Export for Machinery

The approved lines can be exported for:

- tractors with automatic guidance;
- GNSS receivers;
- total stations;
- drones;
- mobile applications.

Each line retains its attributes during export.

6.16 Integration With Sensors

The methodology supports the inclusion of:

- humidity sensors;
- weather stations;
- rain gauges;

- compaction sensors;
- drone images;
- satellites.

These data allow the design to be recalibrated.

6.17 Spatial database

All information must be stored in a spatial database.

Recommended layers:

- DTM;
- slopes;
- drainage;
- floors;
- vegetation;
- LDI lines;
- monitoring;
- field observations.

This makes maintenance and future updates easier.

6.18 Version control

Each design modification must be recorded.

Minimum information:

- date;
- responsible;
- parameters used;
- changes made;
- project version.

It is recommended to use a version control system for scripts and documentation.

6.19 Automation of Quality Control

Before approving a design, the system must automatically verify:

- invalid geometries;
- duplicate lines;
- intersections with drainage systems;
- excessive slopes;
- radii below the minimum;
- isolated segments;
- topological errors.

Only designs that pass these checks will proceed to the export phase.

6.20 Automation through workflows

The methodology can be implemented as an automated processing workflow in which each module generates the inputs for the next. This allows the entire design to be rerun when the DTM, soil data, or constraints are updated, without needing to manually repeat each operation.

6.21 Proposed architecture for the LDI methodology

A five-layer architecture is proposed:

1. **Data layer:**DTM, soils, climate, land use, and constraints.
2. **Analysis layer:**calculation of topographic and hydrological variables.
3. **Design layer:**generation and evaluation of the LDIs.
4. **Optimization layer:**automatic adjustment according to defined criteria.
5. **Presentation layer:**maps, reports, and files for field execution.

This separation facilitates maintenance and the future addition of new functionalities.

6.22 Towards an LDI Platform

As a natural evolution of the methodology, it is proposed to develop a specific platform for the design of Infiltration Lines, with a graphical interface that allows you to:

- create projects;
- import geospatial data;
- run the complete design workflow;
- visualize indicators;
- compare scenarios;
- export results.

The platform could integrate with QGIS through an add-on (plugin) or operate as a standalone application.

6.23 Implementation Roadmap

To facilitate software development, the following order is proposed:

Fase 1

- Python scripts to automate individual tasks.

Fase 2

- Toolbox for QGIS.

Fase 3

- QGIS plugin with a graphical interface.

Fase 4

- Multi-criteria optimization engine.

Fase 5

- Integration with Google Earth Engine for climate, humidity, vegetation, and monitoring data.

Phase 6

- Collaborative web platform for LDI project management.
-

6.24 Conclusions of the Chapter

Automation turns the LDI methodology into a reproducible, scalable, and verifiable design system. By organizing the process into independent modules, it is possible to implement, maintain, and improve the workflow without altering the fundamental principles of the methodology.

The incorporation of explicit rules, automatic evaluation, iterative optimization, and quality control makes it possible to reduce errors and facilitate the execution of projects of any scale.

Proposed Innovations for the Final Version

To further differentiate the LDI methodology from a mere collection of scripts, I propose adding four additional elements to the final version:

1. **Rule Engine:**a configurable component where design criteria (slopes, radii, buffers, constraints) are defined through declarative rules, without modifying the source code.
2. **Optimization Engine:**separate from the GIS analysis, responsible exclusively for finding the best line configuration according to the objective function defined in Deliverable 5.
3. **LDI Data Model:**a standardized spatial database schema (for example, in GeoPackage or PostGIS) with tables for projects, Design Units, lines, evaluations, monitoring, and versions.
4. **Microservices Architecture (future vision):**design the system so that each module (DTM, hydrology, design, optimization, export) can run as an independent service. This will make it easier to integrate with QGIS, Google Earth Engine, web applications, and future tools based on artificial intelligence.

Chapter 7. Field Implementation, Construction, Monitoring, and Continuous Improvement of Infiltration Lines (IDL)

7.1 Introduction

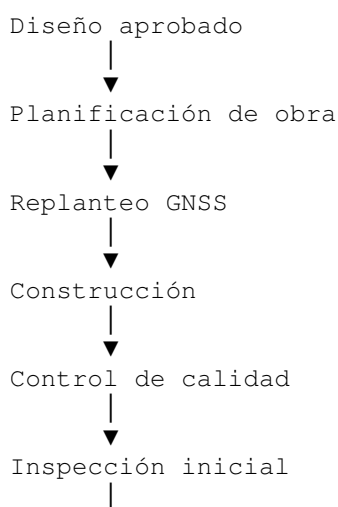
Computational design constitutes only the first stage of the IDL methodology. The project's success depends on the designed lines being built accurately, verified in the field, and subjected to a continuous monitoring and improvement process.

In the IDL methodology, construction does not represent the end of the project, but the beginning of a validation phase in which the system's real behavior is compared with the behavior predicted during design.

The objective of this chapter is to establish a standardized procedure to transfer the digital design to the field, control construction quality, and evaluate the system's hydrological performance throughout its service life.

7.2 Overall Implementation Flow

The complete process comprises nine stages.



▼
First rainfall event |
▼
Mmonitoring
|
▼Mcontinuous
improvement add cod-
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7.3 Construction Planning

Before starting construction, an execution plan must be prepared that includes:

- schedule;
- available machinery;
- operators;
- GNSS system used;
- order of intervention of the Design Units;
- access routes;
- storage areas;
- safety protocols.

Planning aims to minimize unnecessary travel and reduce the impact on the terrain.

7.4 Site Preparation

Before layout is carried out, it must be verified:

- accessibility;
- presence of obstacles;
- fence condition;
- active drainage;
- trees;
- rocks;
- infrastructure;
- soil conditions.

If conditions differ significantly from the digital model, it will be necessary to update the design.

7.5 Reassessment of the lines

The approved lines must be transferred to the field using high-precision equipment.

It is recommended:

- GNSS RTK;
- total stations;
- multi-band receivers;
- agricultural autopilot.

The recommended horizontal and vertical accuracy is less than ± 5 cm.

7.6 Verification prior to construction

Before the machinery enters, it will be checked:

- match between the design and the terrain;
- existence of new drainage systems;
- modifications to the landform;
- presence of recent erosion;
- changes in land use.

Any modification must be recorded.

7.7 Construction of the Lines

Construction must follow strictly the approved layout.

During this phase, the following must be respected:

- working depth;
- line separation;
- turning radii;
- maximum length;
- longitudinal slopes.

Improvised modifications must not be made without documenting them.

7.8 Recommended equipment

The methodology supports different attachments depending on the objective.

Subsoilers

To break up compacted layers.

Chisels

When you want to improve infiltration without investing in the soil.

Special plows

In projects where small infiltration berms are required.

Forest implements

For agroforestry systems.

The choice will depend on the soil type and the project's objective.

7.9 Working Depth

The depth must be defined through prior diagnosis.

Factors:

- compaction;
- texture;

- effective depth;
- presence of hardened horizons;
- soil moisture.

Working deeper does not always improve infiltration.

7.10 Adequate Soil Moisture

Construction should not be carried out:

- with saturated soil;
- during heavy rains;
- on plastic clay.

It is also not recommended to work under extremely dry conditions if this creates compacted blocks.

The moisture content must allow fracturing of the profile without producing sealing.

7.11 Quality Control During Construction

Each day's work must be recorded:

- constructed length;
- operator;
- implement;
- depth;
- speed;
- incidents;
- photographs.

This later makes it possible to reconstruct the project history.

7.12 Geometric Verification

Once the lines have been constructed, it is recommended to take a new sample using GNSS.

Verified:

- position;
- elevation;
- slope;
- continuity;
- radii.

Deviations greater than the tolerance must be corrected.

7.13 Construction Indicators

Each Design Unit receives a report.

Example:

Indicator Value
Executed length
Horizontal error
Vertical error
Average deviation
Maximum deviation
Average depth
Compliance (%)

7.14 First Inspection

It must be carried out immediately after the work is completed.

They are verified:

- interruptions;
- obstructions;
- lateral compaction;
- machinery damage;
- ground stability.

7.15 Evaluation after the first rainfall

The first rainfall constitutes the initial validation of the design.

During this inspection, the following must be recorded:

- visible infiltration;
- accumulations;
- overflows;
- erosion;
- sediments;
- flow continuity;
- critical points.

This information is fundamental for recalibrating the methodology.

7.16 Periodic monitoring

It is recommended to inspect the system:

- at the beginning of the rainy season;
- after extraordinary rains;
- at the end of the rainy season;
- before new agricultural work.

Continuous monitoring makes it possible to detect problems before they worsen.

7.17 Monitoring Variables

The main variables are:

Hydrological

- infiltration;
- runoff;

- soil moisture;
- water retention time.

Geometrical

- deformations;
- loss of continuity;
- sedimentation.

Biological

- vegetal cover;
- roots;
- organic matter;
- Natural regeneration.

Operations

- ease of transit;
- required maintenance;
- compatibility with machinery.

7.18 Recommended Instrumentation

Depending on the project scale, the following may be used:

- humidity sensors;
- penetrometers;
- rain gauges;
- weather stations;
- drones;
- fixed cameras;
- satellite images.

7.19 Performance Indicators

It is proposed to evaluate the system using quantitative indicators.

Hydrological Indicators

- runoff reduction;
- infiltration increase;
- increase in residence time;
- runoff peak reduction.

Soil Indicators

- moisture increase;
- decrease in compaction;
- increase in organic matter;
- structural stability.

Biological Indicators

- increase in coverage;
- root development;
- tree survival;
- biodiversity.

Productive Indicators

- biomass production;
- agricultural yield;
- animal stocking rate;
- system productivity.

7.20 Maintenance

LDIs require preventive maintenance.

Recommended activities:

- clearing blockages;
- repairing interruptions;
- control of localized erosion;
- replenishing vegetation;
- verifying drainage.

The frequency will depend on weather conditions and land use.

7.21 Adaptive management

The methodology adopts an adaptive management approach.

Each rainy season generates new information that must be incorporated into the project.

This allows:

- modifying parameters;
- adjusting spacing;
- redesigning sectors;
- improving future projects.

7.22 Historical database

Each project must maintain a permanent record.

Recommended information:

- original design;
- modifications;
- inspections;
- significant rainfalls;
- photographs;
- measurements;
- repairs;
- production.

This database constitutes the system's technical history.

7.23 System Performance Index (SPI-IDI)

An integrated indicator is proposed to evaluate the project's performance after its construction.

$$IDS = 0.30(H) + 0.25(S) + 0.20(B) + 0.15(P) + 0.10(M)$$

where:

- **H**= hydrological performance;
- **S**= soil condition;
- **B**= biological indicators;
- **P**= productive performance;
- **M**= maintenance status.

Proposed classification:

IDS-LDI	Status
>90	Excellent
80–90	Very good
70–80	Good
60–70	Requires adjustments
<60	Requires redesign

7.24 Scientific Validation

To strengthen the methodology, it is recommended to establish monitoring plots where the following are compared:

- areas with LDI;
- areas without intervention;
- different spacings;
- different implements;
- different types of plant cover.

Variables must be measured over several years to quantify long-term effects.

7.25 Integration With Remote Monitoring

Field observations can be complemented with information obtained through:

- drones;
- satellite images;
- updated digital surface models;

- Connected weather sensors.

This makes it possible to detect changes that are not obvious during on-site inspections.

7.26 Knowledge management

Each project must contribute to improving the methodology.

It is recommended to document:

- lessons learned;
- problems encountered;
- solutions implemented;
- innovations;
- recommendations.

This knowledge should be incorporated into future versions of the methodology.

7.27 LDI project certification (proposal)

As a future development, it is proposed to establish a certification system based on compliance levels.

Level	Requirements
LDI Level I	Validated design and documented construction
LDI Level II	Monitoring for at least one complete hydrological cycle LDI
Level III	Quantitative evidence of improvements in hydrology and soil
LDI Level IV	Multi-year monitoring, adaptive management, and full traceability

This framework would facilitate comparisons between projects and promote the adoption of good practices.

7.28 Chapter Conclusions

The implementation in the field represents the transition between digital design and the system's real-world operation. The LDI methodology considers this phase as an ongoing process of building, verification, monitoring, and improvement, where each rainfall event provides information to refine the design and increase the reliability of future projects.

Proposed Innovations for the Final Version

This chapter introduces several elements that strengthen the methodological approach:

1. **Implementation as a complete life cycle:**the methodology no longer ends with the design and instead encompasses planning, construction, monitoring, and continuous improvement.
2. **System Performance Index (IDS-LDI):**an indicator that makes it possible to objectively assess the operation of the line network once it has been built.
3. **Evidence-based adaptive management:**the project evolves through the systematic incorporation of field data, sensors, and post-rainfall observations.
4. **LDI project certification proposal:**an evaluation framework that could help standardize project quality and facilitate future comparative research.

LDI Methodology V1.0

Computational Design of Infiltration Lines for the Regenerative Hydrologic Management of the Landscape

Version:1.0 (Base Document)

Presentation

This methodology establishes a technical, reproducible, and verifiable procedure for the design of Infiltration Lines (LDI) through the integration of geomorphology, hydrology, hydraulics, soil science, Geographic Information Systems (GIS), mathematical optimization, and artificial intelligence.

The aim is to replace methods based solely on visual interpretation of the terrain with a quantitative, documented procedure that is suitable for automation.

The methodology has been designed for applications in regenerative agriculture, silvopastoral systems, hydrologic restoration, watershed management, and land-use planning.

1. Methodology Philosophy

The LDI methodology starts from a fundamental principle:

Infiltration lines are not designed; the future behavior of water over the landscape is designed.

Infiltration lines are only a tool to modify:

- the flow velocity;
 - the water residence time;
 - infiltration;
 - the spatial distribution of moisture;
 - hydrological connectivity.
-

2. Guiding Principles

All the methodology is based on ten principles.

Principle 1

Relief controls the movement of water.

Principle 2

Infiltration is a priority over drainage.

Principle 3

Each line must serve a specific function.

Principle 4

The design must be based on measurable data.

Principle 5

All designs must be hydraulically verified.

Principle 6

The automation must be transparent.

Principle 7

Monitoring is part of the design.

Principle 8

The Methodology Must Be Continuously Improved.

Principle 9

The soil is a living hydraulic infrastructure.

Principle 10

The final decision always belongs to the designer.

3. General Structure

The complete methodology is organized into seven major components.

I . Scientific Foundations

↓

II . Basic Information

↓

III . Geometric Design

↓

IV . Hydrological Model

↓

V . Automation

↓

VI . Construction

↓

VII . Monitoring

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4. Conceptual Architecture

The LDI system integrates six scientific domains.

ins.Geomorphology

+

Hydrology

+

Hydraulics

+

Soil science

+

GIS

+

Optimization

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None of these components is sufficient on its own. The

methodology arises from its integration.

5. Methodological Flow

Data

↓

DTM

↓

Analysis

↓

Design Units

↓

Mother Line

↓

LDI Generation

↓

Evaluation

↓

Optimization

↓

Construction

↓

Monitoring

↓

Feedback add coderepl-
ace

6. Main Components

Component A

Information.

Includes:

- DTM
 - Grounds
 - Climate
 - Vegetation
 - Restrictions
-

Component B

Analysis.

Calculate:

- slope;
 - curvatures;
 - accumulation;
 - moisture;
 - connectivity.
-

Component C

Design.

Generate:

- Main Line;
 - derived lines;
 - functional classification.
-

Component D

Evaluation.

Calculate:

- ICL;
 - ICI;
 - ICH;
 - IG-LDI.
-

Component E

Construction.

Includes:

- re-measurement;
- execution;
- quality control.

Component F

Monitoring.

Evaluates:

- infiltration;
- humidity;
- erosion;
- productivity.

7. Conceptual model

Every decision is summarized in a logical chain.

Relief

↓

Water movement

↓

Hydraulic behavior

↓

Infiltration

↓

Moisture

↓

Vegetation

↓

Production

↓

Resilience

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8. Decision Levels

The methodology operates simultaneously at four scales.

Strategic

Basin.

Tactical

Farm.

Operational

Design Unit.

Executive

Each line.

9. Project Life Cycle

Design

↓

Construction

↓

Monitoring

↓

Calibration

↓

Redesign

↓

New version

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The methodology considers that a project is never completely finished.

10. Computer architecture

The proposed software will be divided into nine engines.

DTM Engine

Construction.

Hydrological Engine

Variables.

Geomorphological Engine

Classification.

LDI Engine

Design.

Hydraulic Engine

Evaluation.

Optimization Engine

Correction.

AI Engine

Learning.

SIG Engine

Visualization.

Reporting Engine

Export.

11. Global Indicators

The methodology proposes five indicators.

Indicator	Objective
ICL Line Quality	
ICI Infiltration Capacity	ICH Hydr-
ological Connectivity	
IG-LDI Overall design quality	
IDS Performance after construction	

12. Generated Products

Each project automatically produces:

- maps;
 - curves;
 - profiles;
 - tables;
 - indicators;
 - GIS files;
 - GNSS files;
 - technical reports;
 - history.
-

13. Methodological innovations

Regarding traditional methods, the LDI methodology incorporates:

13.1 Design Units (DU)

The terrain is no longer treated as a uniform surface.

13.2 Smart Mother Line

It is not based solely on contour lines.

Integrates:

- hydrology;
 - geomorphology;
 - constraints.
-

13.3 Variable spacing

Each line has a different spacing.

13.4 Functional Classification

The lines can be:

- infiltration;

- redistribution;
 - dissipation;
 - agroforestry;
 - protection.
-

13.5 Automatic evaluation

Each line receives a score.

13.6 Iterative optimization

The design automatically improves.

13.7 Continuous monitoring

Construction does not represent the end.

13.8 Hydrological Digital Twin

All historical information remains associated with the project.

14. Vision for the Future

The methodology has been designed to evolve into an integrated platform capable of integrating:

- IoT sensors;
- drones;
- satellites;
- weather stations;
- artificial intelligence;
- machine learning;
- hydraulic simulation;
- precision agriculture.

15. Technology roadmap

Stage 1

Methodological manual.

(this document)

Stage 2

Python scripts.

Stage 3

QGIS toolbox.

Stage 4

QGIS plugin.

Stage 5

Web Application.

Stage 6

LDI Platform.

Stage 7

Digital Twin.

16. Future research lines

They are identified as priorities:

- relationship between line spacing and soil texture;
 - integration with weather forecasts;
 - use of soil moisture models;
 - economic evaluation;
 - carbon capture;
 - biodiversity;
 - climate resilience;
 - optimization through evolutionary algorithms;
 - calibration with sensors in real time.
-

17. General conclusions

The LDI methodology proposes a paradigm shift in the design of infiltration infrastructure. Rather than reproducing uniform geometric patterns, it proposes a data-driven adaptive process in which each decision is based on measurable variables and validated through objective indicators.

The combination of geomorphological analysis, hydrological modeling, computational optimization, and continuous monitoring makes it possible to develop reproducible and scalable designs, suitable for both site-specific projects and large-scale hydrological restoration programs.
